INCH-POUND

MIL-PRF-22710/24B
20 April 2001

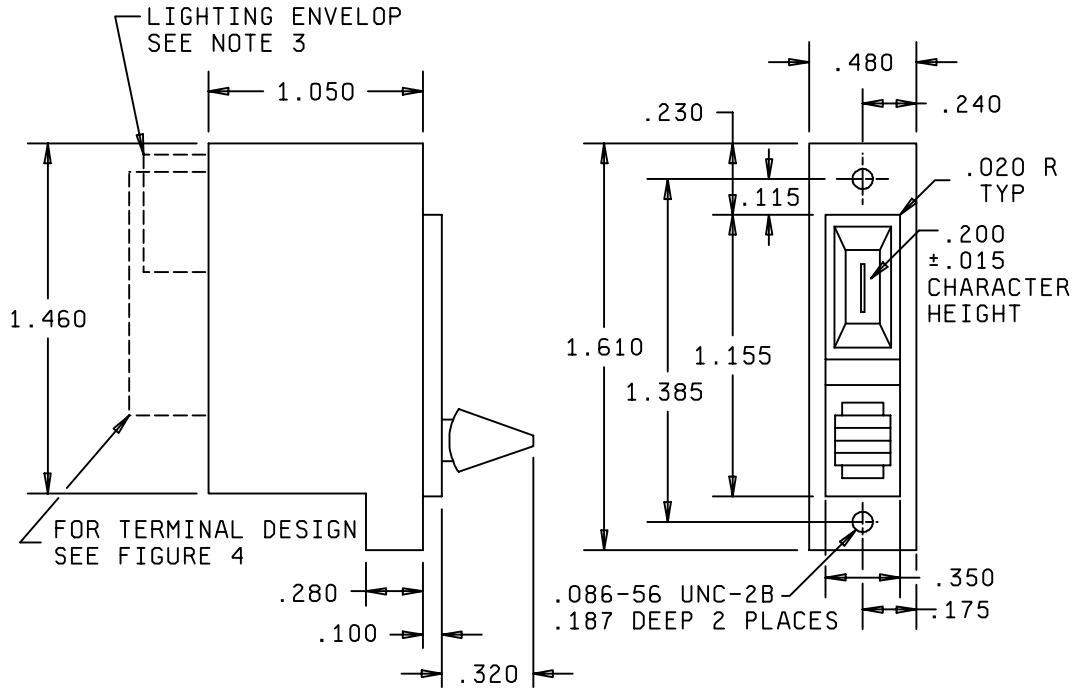
SUPERSEDING
MIL-S-22710/24A
14 March 1977

PERFORMANCE SPECIFICATION SHEET

SWITCHES, ROTARY (PRINTED CIRCUIT), (TOGGLE, BIDIRECTIONAL, SEALED, ILLUMINATED OR NONILLUMINATED), STYLE SRPC24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-PRF-22710 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.

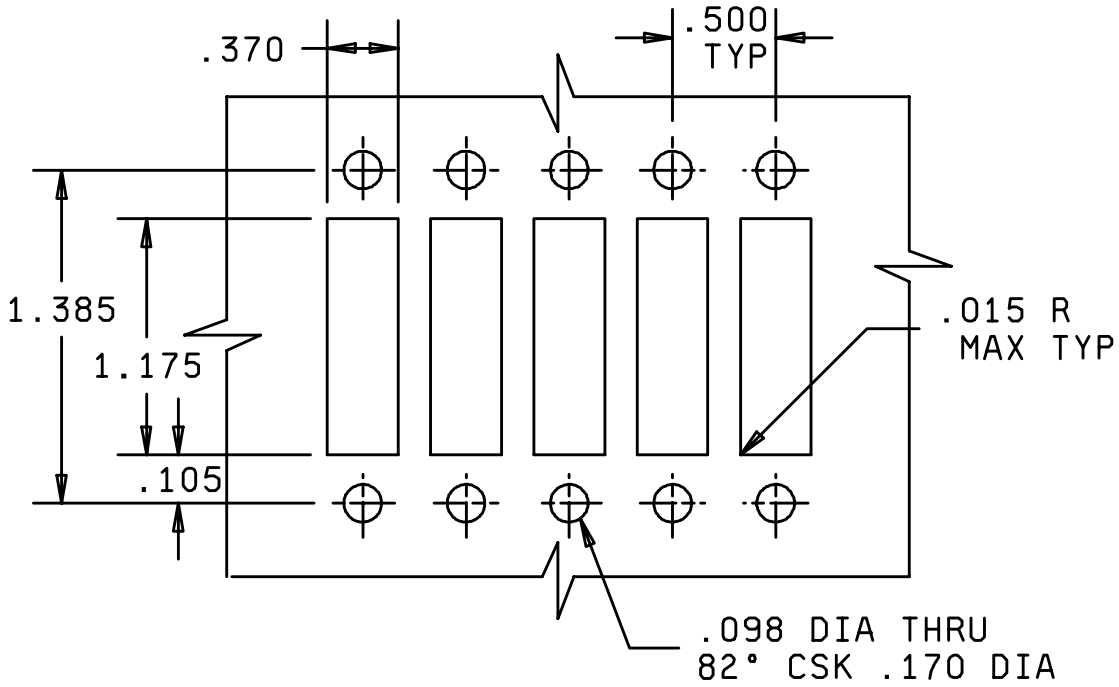


INCHES	MM	INCHES	MM	INCHES	MM
0.015	0.38	0.180	4.57	0.350	8.89
0.020	0.51	0.200	5.08	0.480	12.19
0.086	2.18	0.230	5.84	1.050	26.67
0.100	2.54	0.240	6.10	1.155	29.34
0.115	2.92	0.280	7.11	1.385	35.18
0.175	4.45	0.320	8.13	1.460	37.08
				1.610	40.89

FIGURE 1. Style SRPC24 switch and panel cutout and installation data.

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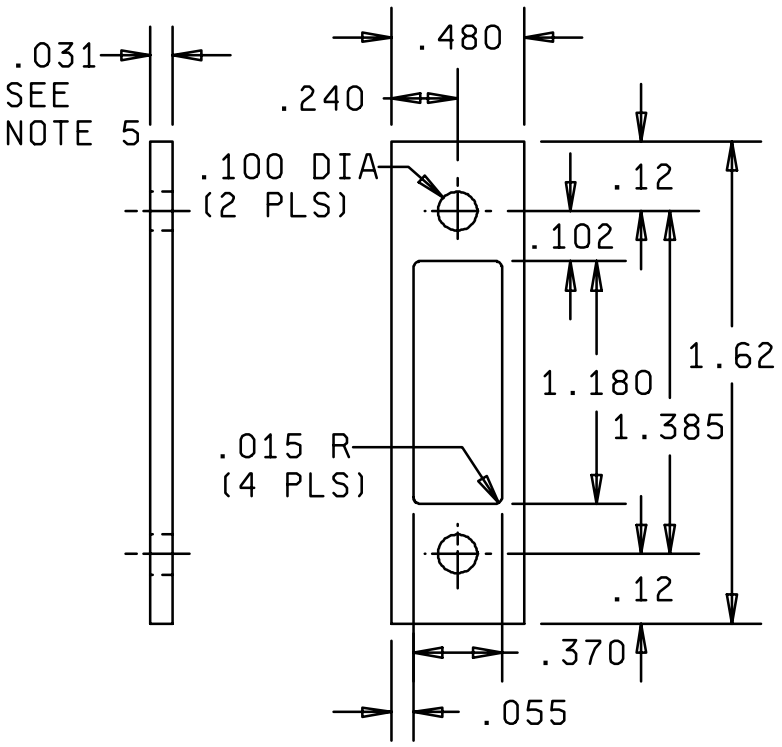


INCHES	MM
0.015	0.38
0.098	2.49
0.105	2.67
0.170	4.32
0.370	9.40
0.500	12.70
1.175	29.85
1.385	35.18

NOTES
1. Dimensions are in inches. Metric equivalents are for general information only.
2. Unless otherwise specified, tolerances are ± .010 (.25 mm).
3. Applicable to illuminated switches only. Allowance for lighting terminals for externally replaceable lamps, when internal lighting is specified.

FIGURE 1. Style SRPC24 switch and panel cutout and installation data - Continued.

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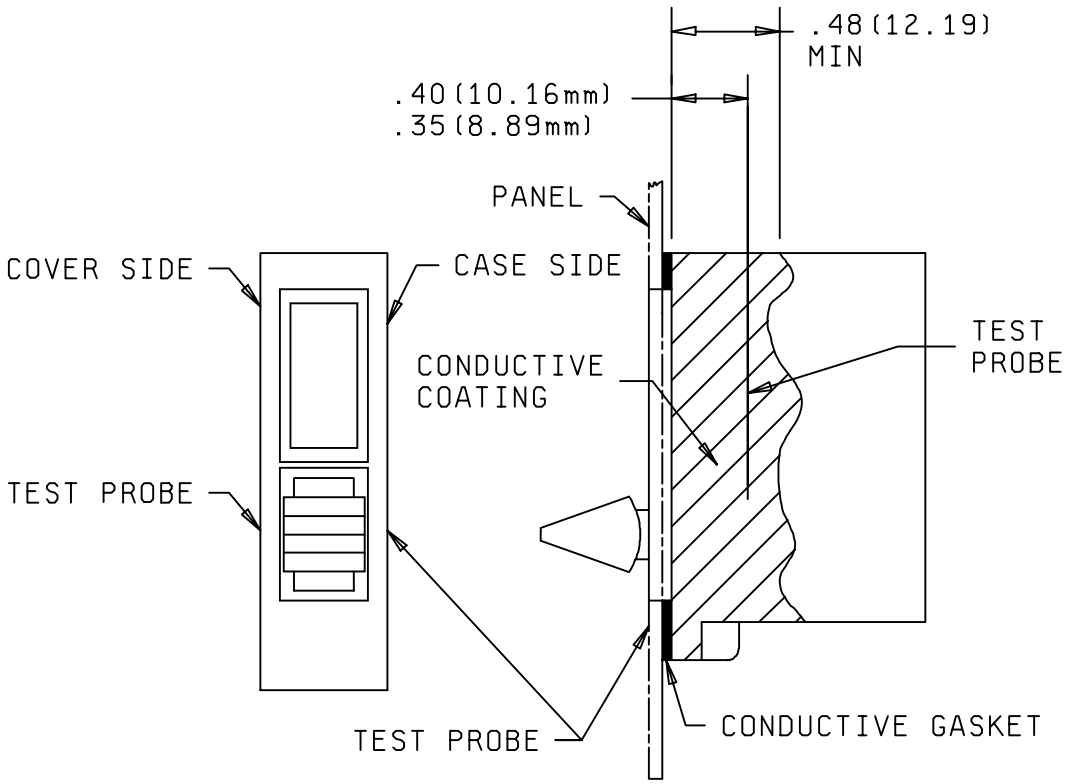


INCHES	MM	INCHES	MM
0.015	0.38	0.240	6.10
0.031	0.79	0.370	9.40
0.055	1.40	0.480	12.19
0.100	2.54	1.180	29.97
0.102	2.59	1.385	35.18
0.120	3.05	1.625	41.28

NOTES
1. Dimensions are in inches. Metric equivalents are for general information only.
2. Unless otherwise specified, tolerances are ± .005 (.13 mm) for three place decimals and ±..02 (.5 mm) for two place decimals.

FIGURE 2. Panel seal gasket.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based on 1 inch = 25.4 mm.

FIGURE 3. RFI shielded switch.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2 and 3.

Construction: D.

Angle of Throw: 36°

Temperature-life characteristic: B, except -55° low temperature for all switches and +71° C for lighted switches.

TABLE I. Electrical test load.

Test of condition	Resistive	
At atmospheric pressure	0.125 A	28 V dc and ac, 60 Hz
At reduced barometric pressure	0.125 A	28 V dc and ac, 60 Hz
Life (low level)	10 mA	30 mV

Vibration grade: 3 (15g, 10 - 2,000 Hz).

Shock types: M (100 g's) and H (high-impact). Methods I and II are applicable.

Altitude rating: C (70,000 feet).

Terminal design: See figure 3 and table III.

Contact resistance:
Initial: 50 milliohms, maximum.
Following life (rotational): 70 milliohms, maximum.

Luminace: Illuminated switches only.

Moisture resistance: Method 103, test condition A, MIL-STD-202, with polarization voltage.

TABLE II. Photometric brightness.

Lamp	Switch filter	Brightness	Rating	
			Volts	Ampere
M6363/6 or equivalent	Clear	2.0 ± 1.0 foot-lamberts	5	0.115
	Red	1.0 ± 0.5 foot-lamberts	5	0.115
	Amber	2.0 ± 1.0 foot-lamberts	5	0.115

Operating force (tangential): 12 ounces minimum
36 ounces maximum

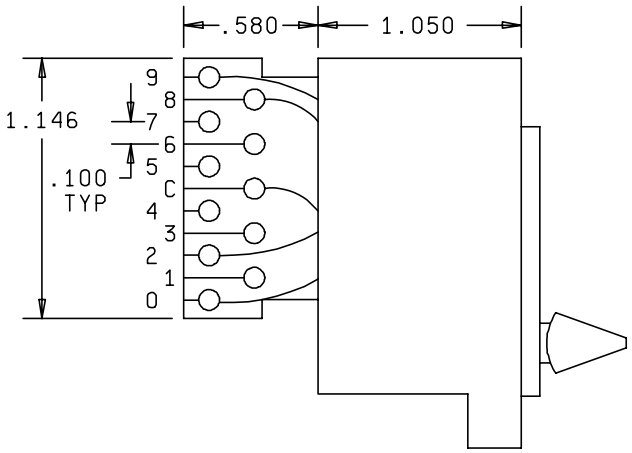
Seal: Method II, when specified in table I.

Dielectric withstanding voltage: 750 volts rms at atmospheric pressure
250 volts rms at reduced barometric pressure, test condition C.

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EMI/RFI shielding: Shielded switches (see table III) are intended for use in equipment required to meet the radiated (R) portions of MIL-STD-461. The conductive coating and conductive gasket shall be as shown in figure 3. When the switch is mounted facing outward to an electrically conductive panel, the resistance shall be measured between the RFI shielding on the switch and the mounting panel. The resistance shall not exceed 1 ohm.

Military part number: M22710/24- (dash number from table III).



Truth table 1/ Straight decimal	
Dial Marking	Common c Conn to:
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

INCHES	MM
.100	2.54
.580	14.73
1.050	26.67
1.146	29.11

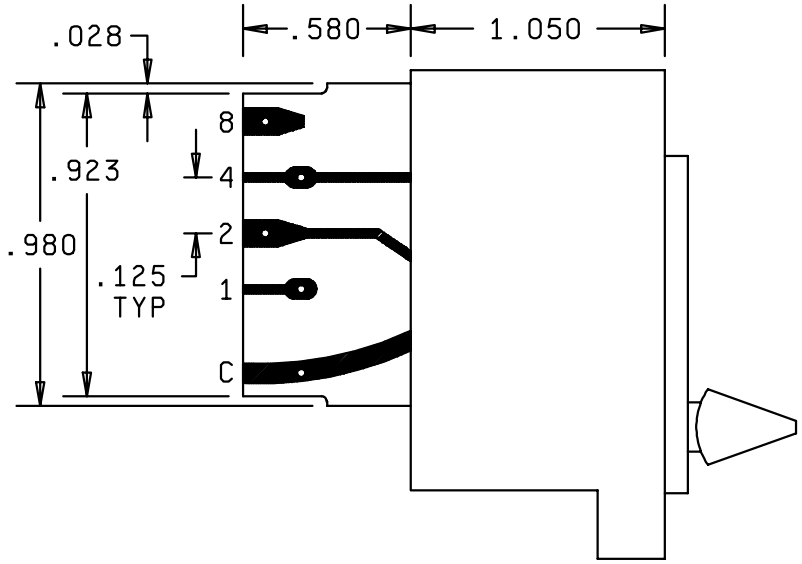
1/ Code explanation: Dial reads 0 - Common C active with terminal 0 only.
Dial reads 1 - Common C active with terminal 1 only.
Dial reads 2 - Common C active with terminal 2 only.
and so forth.

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are base upon 1 inch = 25.4 mm.

Circuit configuration 1

FIGURE 4. Module and circuit configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:34Z
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Truth table 1/ Binary coded decimal				
Dial marking	Common C Conn to:			
	1	2	4	8
0				
1	•			
2		•		
3	•	•		
4			•	
5	•		•	
6		•	•	
7	•	•	•	
8				•
9	•			•

INCHES	MM
.028	.71
.125	3.18
.580	14.73
.923	23.44
.980	24.89
1.050	26.67

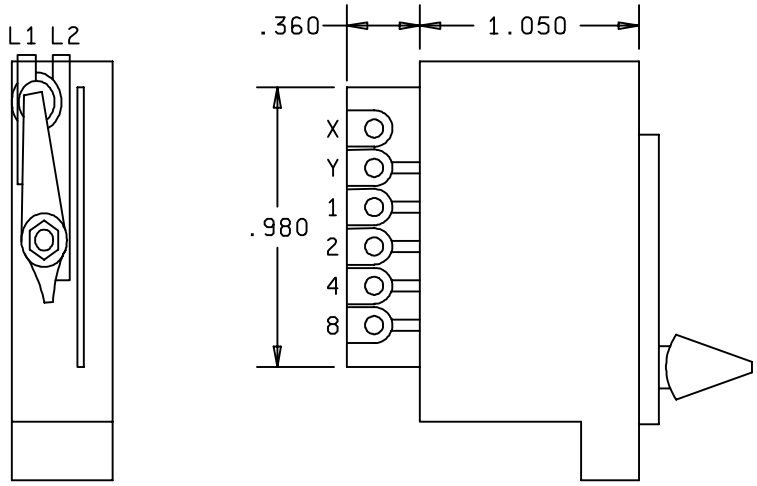
1/ Code explanation: Dial reads 0 - All terminals non-active.
Dial reads 1 - Common C active with terminal 1 only.
Dial reads 2 - Common C active with terminal 2 only.
and so forth.

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are base upon 1 inch = 25.4 mm.

Circuit configuration 2

FIGURE 4. Module and circuit configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:34Z
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Truth table 1/ Binary coded octal with two commons				
Dial marking	Common X(•) Y(o) Conn to:			
	1	2	4	8
0	o	o	o	o
1	•	o	o	o
2	o	•	o	o
3	•	•	o	o
4	o	o	•	o
5	•	o	•	o
6	o	•	•	o
7	•	•	•	o
8	o	o	o	•
9	•	o	o	•

INCHES	MM
.360	9.14
.980	24.89
1.050	26.67

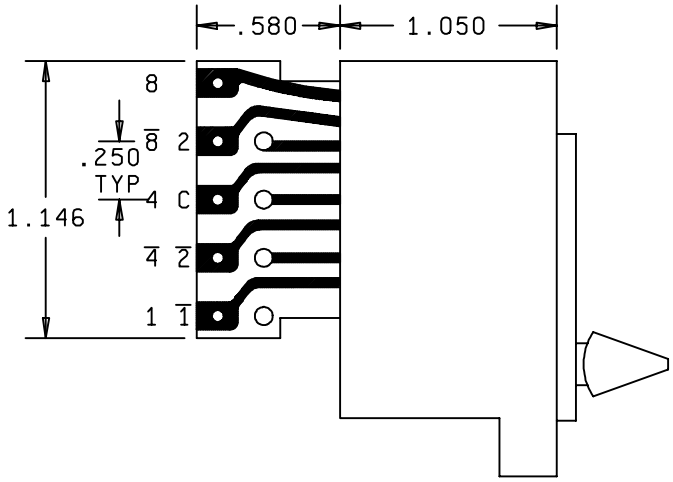
1/ Code explanation: Dial reads 0 - Common Y active with terminals 1, 2, 4 and 8 while common X is not connected to any terminals.
Dial reads 1 - Common X connected to terminal 1, while common Y is connected to terminals 2, 4 and 8.
Dial reads 2 - Common X connected to terminal 2, while common Y is connected to terminals 1, 4 and 8.
and so forth.

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are base upon 1 inch = 25.4 mm.
3. Where lighting is specified, terminals shall be marked L1, L2.

Circuit configuration 3

FIGURE 4. Module and circuit configuration - Continued.

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Truth table 1/ Binary coded decimal plus compliment								
Dial marking	Common C Conn to:							
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					•	•	•	•
1	•					•	•	•
2		•			•		•	•
3	•	•					•	•
4			•		•	•		•
5	•		•			•		•
6		•	•		•			•
7	•	•	•					•
8				•	•	•	•	
9	•			•		•	•	

INCHES	MM
.250	6.35
.580	14.73
1.050	26.67
1.146	29.11

1/ Code explanation: Dial reads 0 - Common C active with terminals $\overline{1}$, $\overline{2}$, $\overline{4}$ and $\overline{8}$
Dial reads 1 - Common C active with terminals 1, $\overline{2}$, $\overline{4}$ and $\overline{8}$.
Dial reads 2 - Common C active with terminals 2, 1, $\overline{4}$ and $\overline{8}$.
and so forth.

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are base upon 1 inch = 25.4 mm.

Circuit configuration 4

FIGURE 4. Module and circuit configuration - Continued.

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Table III. Part numbers and characteristics.

Part Number M22710/24-	Circuit configuration	Dial		illumination		Special features
		Background	Marking	Color	Location	
001	1	Black Opaque	White Opaque	---	---	---
002	1	Black Opaque	White Opaque	---	---	RFI Shielded
003	1	Black Opaque	White Translucent	Red	Back	---
004	1	Black Opaque	White Translucent	Red	Back	RFI Shielded
005	1	Black Opaque	White Translucent	Clear	Back	---
006	1	Black Opaque	White Translucent	Clear	Back	RFI Shielded
007	2	Black Opaque	White Opaque	---	---	---
008	2	Black Opaque	White Opaque	---	---	RFI Shielded
009	2	Black Opaque	White Translucent	Red	Back	---
010	2	Black Opaque	White Translucent	Red	Back	RFI Shielded
011	2	Black Opaque	White Translucent	Clear	Back	---
012	2	Black Opaque	White Translucent	Clear	Back	RFI Shielded
013	3	Black Opaque	White Opaque	---	---	---
014	3	Black Opaque	White Opaque	---	---	RFI Shielded
015	3	Black Opaque	White Translucent	Red	Back	---
016	3	Black Opaque	White Translucent	Red	Back	RFI Shielded
017	3	Black Opaque	White Translucent	Clear	Back	---
018	3	Black Opaque	White Translucent	Clear	Back	RFI Shielded
019	3	Black Opaque	White Translucent	Amber	Back	---
020	3	Black Opaque	White Translucent	Amber	Back	RFI Shielded
021	4	Black Opaque	White Opaque	---	---	---
022	4	Black Opaque	White Opaque	---	---	RFI Shielded
023	4	Black Opaque	White Translucent	Red	Back	---
024	4	Black Opaque	White Translucent	Red	Back	RFI Shielded
025	4	Black Opaque	White Translucent	Clear	Back	---
026	4	Black Opaque	White Translucent	Clear	Back	RFI Shielded

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Army - CR
Navy- EC
Air Force - 11
DLA - CC

Review activities:
Army - AR. CR4, MI, SM
Navy - MC
Air Force - 99

Preparing Activity:
DLA - CC

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